

Proposal 1: DevReady – An AI-Powered Technical Interview Simulator

1. Introduction

Securing positions at leading technology companies requires not only strong technical knowledge but also the ability to perform effectively under interview pressure an ability that is rarely addressed in traditional academic curricula. This project proposes **DevReady**, a web-based platform designed to simulate realistic technical interviews using an AI-powered interviewer. The system evaluates users across data structures and algorithms (DSA), system design thinking, and behavioral responses, while providing structured, actionable feedback. DevReady tracks performance over time and generates personalized improvement suggestions. From a software engineering perspective, the project involves integrating large language model (LLM) based evaluation, speech-to-text processing, in-browser code execution, and performance analytics, resulting in a production-ready platform with clear startup potential.

2. Statement of the Problem

Many technically capable software engineering students struggle to translate their skills into successful interview outcomes. One major issue is the lack of realistic interview practice environments that replicate real-world pressure, time constraints, and communication expectations. Students typically prepare in isolation by solving problems or reading materials, without experiencing structured interviews. Additionally, there is an absence of objective and consistent feedback, making it difficult for candidates to identify weaknesses in their problem-solving approach or communication style. Professional interview coaching is often expensive and inaccessible, creating inequality among candidates. Furthermore, students lack systematic mechanisms to track performance over time and identify recurring weak areas across multiple interview attempts.

3. Objectives

General Objective:

To design, develop, and deploy an AI-powered interview simulation platform that helps

software engineering students improve interview readiness, technical communication, and confidence through realistic practice and performance tracking.

Specific Objectives:

- To develop an AI-powered interview engine by integrating an LLM API to conduct adaptive interviews covering DSA, system design, and behavioral questions with context-aware follow-up prompts.
- To build an in-browser code evaluation environment that allows users to write, execute, and submit solutions during interviews, with automated correctness and efficiency assessment.
- To implement speech-to-text processing for verbal responses, enabling analysis of communication clarity, confidence, and appropriate use of technical terminology.
- To generate detailed AI-driven feedback after each interview session, highlighting strengths, weaknesses, and targeted improvement recommendations.
- To design a performance analytics dashboard that tracks user progress over time, visualizes trends, identifies weak areas, and suggests focused practice topics.

4. Significance of the Project

This project offers strong academic and practical value. From an engineering standpoint, it exposes the development team to advanced areas such as LLM prompt engineering, secure and sandboxed code execution, speech processing, and data visualization skills that are highly relevant in today's software industry. From a practical perspective, DevReady addresses a real and widespread challenge faced by software engineering students preparing for competitive technical interviews. By providing accessible, objective, and scalable AI-driven interview practice, the platform transforms interview preparation into a structured, measurable, and guided experience. Its focused scope and clear value proposition position it as a potentially impactful product within the developer tools ecosystem.

Proposal 2: HelpMeStudy – An AI-Powered Personal Study Assistant

1. Introduction

Managing and understanding large volumes of study material is one of the most persistent challenges faced by students in higher education. This project proposes **HelpMeStudy**, a web-based intelligent study assistant that enables students to upload study materials in multiple formats, including PDF files, PowerPoint slides, Word documents, and images of handwritten notes. The system uses artificial intelligence to generate clear explanations, auto-create quizzes, and provide personalized follow-up recommendations. By tracking performance across study sessions, StudyMind offers analytics that help students identify weak areas and improve efficiently. From a software engineering perspective, the project involves multi-format file processing, large language model (LLM)–based content understanding, adaptive quiz generation, and performance analytics, resulting in a practical and impactful learning support platform.

2. Statement of the Problem

Despite having access to lecture notes, textbooks, and digital resources, many students struggle to study effectively. Large volumes of unstructured material often overwhelm learners, making it difficult to extract and understand key concepts efficiently. Traditional study methods are largely passive and do not adapt to individual learning needs, leaving students unaware of which topics they have mastered and which require more attention. In addition, academic support is not always available outside classroom hours, limiting access to immediate explanations when students encounter difficult concepts. Students also lack effective self-assessment tools to test their understanding of specific materials and track progress over time, reducing the effectiveness of exam preparation.

3. Objectives

General Objective:

To design, develop, and deploy an AI-powered personal study assistant that helps students understand their study materials, assess their knowledge through adaptive quizzes, and track academic progress over time.

Specific Objectives:

- To develop a multi-format content ingestion system that supports uploading and processing PDFs, Word documents, PowerPoint slides, and images of handwritten notes.
- To implement an AI-powered explanation engine using an LLM API to generate clear, contextual explanations of uploaded study materials and support follow-up questions.
- To design an adaptive quiz generation module that automatically creates multiple-choice, true/false, and short-answer questions with instant grading and feedback.
- To build a personalized follow-up recommendation system that analyzes quiz performance to identify weak areas and suggest targeted revision using spaced repetition principles.
- To develop a performance analytics dashboard that visualizes quiz scores, study history, topic-level progress, and personalized improvement recommendations.

4. Significance of the Project

This project provides significant value both academically and practically. From a software engineering standpoint, it offers hands-on experience in file processing, LLM prompt engineering, adaptive assessment logic, database design, and data visualization within a cohesive system. Practically, StudyMind transforms passive studying into an interactive, personalized, and measurable learning experience. It enables students to access on-demand academic support, focus on weak areas through targeted practice, and improve knowledge retention over time. With its broad applicability and intuitive design, the system has strong potential for real-world deployment and future expansion within the educational technology domain.

Proposal 3: QR Code–Based Student Attendance Management System

1. Introduction

Accurate and efficient attendance tracking is a critical component of academic administration, yet many universities still rely on manual methods such as paper sign-in sheets and roll calls that are time-consuming and error-prone. This project proposes a **QR Code–Based Student Attendance Management System**, a web-based solution that automates attendance recording by generating a unique QR code for each class session. Students scan the QR code using their devices to instantly mark their attendance, while lecturers can monitor, manage, and export attendance reports from a dedicated dashboard. The project requires the implementation of secure authentication, dynamic QR code generation, real-time attendance recording, and a comprehensive reporting system, resulting in a practical and deployable solution for academic institutions.

2. Statement of the Problem

Academic institutions face persistent challenges with traditional attendance management practices that reduce efficiency and accountability. Roll calls and paper-based sign-in sheets consume valuable class time and increase the administrative workload of lecturers, disrupting teaching sessions. Manual attendance recording is prone to human error, including incorrect entries, duplicate records, and misplaced attendance sheets, leading to unreliable data that can affect student academic standing. In addition, proxy attendance where students sign in on behalf of absent peers is difficult to prevent using manual systems. Generating meaningful attendance reports or tracking attendance trends over time is also labor-intensive, limiting effective academic and administrative oversight.

3. Objectives

General Objective:

To design and develop a web-based QR code–based attendance management system that enables lecturers to record and manage student attendance efficiently, accurately, and transparently.

Specific Objectives:

- To develop a secure authentication system by implementing role-based access control with separate portals for lecturers, students, and administrators.
- To implement dynamic QR code generation that produces unique, time-limited QR codes for each class session to reduce fraudulent or proxy attendance.
- To enable real-time attendance recording by allowing students to scan session QR codes using their personal devices.
- To build an attendance management module that provides lecturers with dashboards to view, edit, and manage attendance records by course, session, and student.
- To generate comprehensive attendance reports that can be filtered by course, date range, and student, with export options in PDF and Excel formats.

4. Significance of the Project

This project offers strong academic and practical value. From a software engineering perspective, it provides hands-on experience in QR code generation, real-time data processing, role-based authentication, database management, and report generation within a well-defined and achievable scope. Practically, the system reduces time spent on manual attendance taking, minimizes errors and proxy attendance, and provides institutions with accurate and easily accessible attendance data. By improving transparency and accountability for both students and lecturers, the project contributes to the digital transformation of academic administration and represents a solution that can be readily deployed in real university environments.